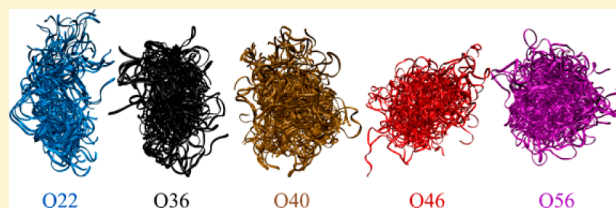


Emerging β -Sheet Rich Conformations in Supercompact Huntingtin Exon-1 Mutant StructuresHong Suk Kang,[†] Francisco X. Vázquez,[†] Leili Zhang,[†] Payel Das,[†] Leticia Toledo-Sherman,[‡] Binquan Luan,[†] Michael Levitt,[§] and Ruhong Zhou^{*,†,||}[†]Computational Biology Center, IBM Thomas J. Watson Research Center, Yorktown Heights, New York 10598, United States[‡]CHDI Management/CHDI Foundation, Los Angeles, California 90045, United States[§]Department of Structural Biology, Stanford University School of Medicine, Stanford, California 94305, United States^{||}Department of Chemistry, Columbia University, New York, New York 10027, United States

S Supporting Information

ABSTRACT: There exists strong correlation between the extended polyglutamines (polyQ) within exon-1 of Huntingtin protein (Htt) and age onset of Huntington's disease (HD); however, the underlying molecular mechanism is still poorly understood. Here we apply extensive molecular dynamics simulations to study the folding of Htt-exon-1 across five different polyQ-lengths. We find an increase in secondary structure motifs at longer Q-lengths, including β -sheet content that seems to contribute to the formation of increasingly compact structures. More strikingly, these longer Q-lengths adopt supercompact structures as evidenced by a surprisingly small power-law scaling exponent (0.22) between the radius-of-gyration and Q-length that is substantially below expected values for compact globule structures (~ 0.33) and unstructured proteins (~ 0.50). Hydrogen bond analyses further revealed that the supercompact behavior of polyQ is mainly due to the "glue-like" behavior of glutamine's side chains with significantly more side chain-side chain H-bonds than regular proteins in the Protein Data Bank (PDB). The orientation of the glutamine side chains also tend to be "buried" inside, explaining why polyQ domains are insoluble on their own.



■ INTRODUCTION

Huntington's disease (HD) is caused by the expansion of nucleotide CAG repeats in exon-1 of the HD gene, a mutation that encodes an elongated polyglutamine tract (polyQ) within the N-terminal region of the Huntingtin (Htt) exon-1 protein.¹ In adult-onset HD, the pathogenic threshold polyQ length is above 36 glutamine (Q) repeats.^{1,2} Although a negative correlation between polyQ length and the onset age of HD symptoms has been well established,^{2–4} the mechanisms by which extended polyQ regions cause neuronal dysfunction and cell death remain unclear. Mutant Htt (mtHtt) variants (here we call Q-lengths above 36 as mutants, and Q22 as the "wild-type" wtHtt; more below) are known to form oligomers that aggregate into large, insoluble protein complexes (such as amyloid fibrils) that were previously believed to constitute the principle source of toxicity in HD.⁵ Recent in situ immunocytochemistry assays, however, have revealed that monomeric and lightweight oligomeric forms of mtHtt are also associated with neuronal death.⁶ As these experimental results emphasize, elucidating the relationship between polyQ length and protein structure is crucial for understanding the driving forces behind mtHtt oligomer formation and identifying the processes by which these oligomers and larger aggregates lead to neural toxicity.

The Htt exon-1 protein is intrinsically disordered and, as a direct consequence, difficult to characterize structurally. Exon-1 monomers can access a range of structures at physiological temperatures, and it is not known which of these configurations form the basis for toxicity. Previous studies have focused on determining the structures of individual exon-1 domains, including (1) its 17-residue N-terminal region (N17), (2) its polyQ tract, and (3) its 38-residue, polyproline-rich segment near its C-terminus (C38). Experimentally determined structures suggest that N17 folds into an amphipathic α -helix when it is attached to protein tags or placed in a micellar environment.^{7–9} Nuclear magnetic resonance (NMR) measurements, however, indicate that N17 is intrinsically disordered in solution;¹⁰ however, it has been suggested the N17 helices in solution may aggregate to initialize polyQ aggregation.¹¹ The polyQ region is also intrinsically disordered in solution, but collapses into a compact globule when both N17 and C38 are absent and the Q-length is 20 or greater.^{5,12–16} Notably, the polyQ region within an Htt36Q3H variant of Htt (polyQ = Q₇HQHQHQ₇) was found to adopt β -hairpin configurations, which may play a role in the formation of mtHtt fibrils.^{15,17}

Received: January 24, 2017

Published: June 13, 2017

what?

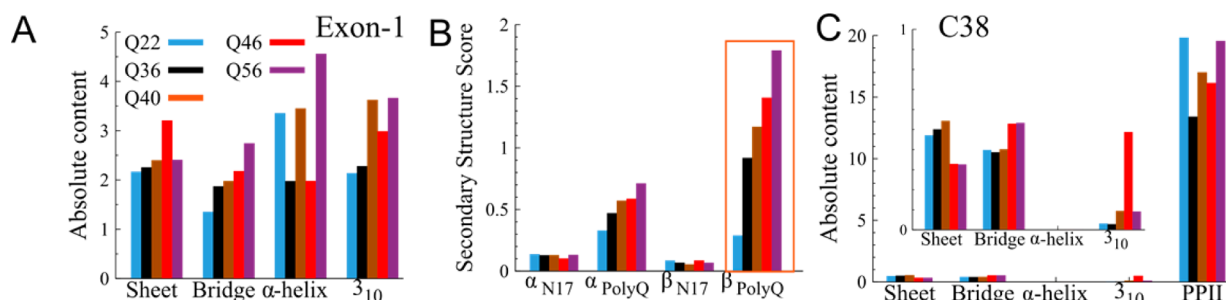


Figure 1. Secondary structural analysis shows that the β -sheet content of Huntingtin (Htt) increases with Q-length faster than the α -helical content. (A) Absolute secondary structure content of the full Htt exon-1 for Q22, Q36, Q40, Q46, and Q56 (B) The average secondary structure scores of first 17 residues near N-terminus (N17) and polyglutamine (polyQ) regions of Htt for all Q-lengths. A sharp increase in β -sheet content emerges between Q22 and the pathogenic threshold Q-length, Q36. The estimated errors are less than 3%. (C) Absolute secondary structure content of the proline-rich segment near C-terminus (C38) for all Q-lengths. Because the content of the polyproline II (PPII) helices is so high, the inset shows other secondary structural motifs on a reduced scale.

Molecular dynamics (MD) simulations have well complemented experimental studies by sampling ensembles of possible intermediate states,^{18–20} a scheme that is particularly crucial for understanding the thermodynamics of intrinsically disordered proteins (IDPs). In this regard, computational studies have provided fruitful insight into the structures of N17 and various polyQ domains.^{21–24} However, due to the intrinsic complexity of these large IDPs and limitations on computational resources, these early simulation efforts have yielded mixed results (despite numerous attempts with different simulation techniques involving discrete molecular dynamics engines, implicit solvents, low resolution coarse-grained (CG) models, etc.). Some of these computational studies have suggested that α -helices dominate the secondary structural profile of monomeric Htt exon-1,^{21–23} while others have reported significant β -sheet formation in the monomer that increases with Q-length.²⁴ These contradictory observations highlight the need for more extensive simulations conducted with higher resolution models, such as fully atomistic protein representations simulated in explicit solvent.

In this article, we present the results from all-atom, explicit solvent MD simulations of the “wild type” (Q22) and representative “mutant” (Q36, Q40, Q46, and Q56) Htt proteins. To effectively capture the many configurations these IDPs adopt, we carried out extensive temperature replica exchange molecular dynamics (T-REMD) simulations across 128 replicas for at least 500 ns per replica, a technique that substantially improves sampling diversity while maintaining thermodynamic rigor. By leveraging this enhanced sampling to estimate rugged conformational free energy landscapes over a large set of Q-lengths, we are able to codify the Q-length-dependent structural and polymeric properties of Htt exon-1. Intriguingly, we find that the propensity to form β -sheet content increases with increasing Q-length, suggesting that mtHtt oligomerization and fibril formation might be influenced by this trend. This increase in β -sheet content within the polyQ region leads to a decrease in the number of hydrogen bonds between the N17 and C38 domains, an observation consistent with recent FRET experiments.²⁵ Our analysis of the polymeric properties of Htt reveals that the wild type Htt (wtHtt) is more flexible than mtHtts, which favor more condensed structures. The scaling behavior for the polyQ segment’s radius of gyration (R_g) implies that at high Q-lengths, the polyQ region adopts supercompact structures not seen in other globular proteins. This increased compactness at long Q-lengths suggests that

extended polyQ domains may lead to increased neurotoxicity both by inducing distinct morphological changes throughout the entire exon-1 protein. One may also suspect that the augmented β -content within and suppressed radial extension of the polyQ domain would impact Htt’s interactions with itself, nearby exon-1 monomers, and its distinct protein binding partners.

RESULTS

Secondary Structural Analysis. Experimental studies of polyQ chain aggregation have suggested that monomeric proteins with β -sheet content associated with longer Q-lengths may play a role in fibril formation.^{14,26} Artificially constructed β -sheet motifs in polyQ domains have also been shown to increase the rate of polyQ aggregation,²⁷ suggesting that the appearance of β -sheet structures in the polyQ domain represents a key step for the formation of exon-1 fibrils. Here, we have indeed found that the amount of β -sheet content in exon-1 increases with Q-length, as Figure 1A illustrates. The plot shows that, in general, all of the mtHtts ($Q \geq 36$) adopt structures with more residues in β -sheet conformations than are seen in the wtHtt (Q22) structural ensemble. At the extreme, approximately twice as many residues assume β -sheet conformations in the Q46 mutant as compared to the wtHtt. Helical structures are more abundant in the mtHtts, as well.

Because of the transient nature of secondary structure in IDPs, we employed the continuous secondary structure scoring functions developed by Pietrucci and Laio in our analysis.²⁸ These functions assign a continuous β -sheet or α -helix score based on the root-mean-square deviation between the given structure and an ideal structure determined from the CATH database (<http://www.cathdb.info/>). A value of 1.0 represents a sliding segment of six residues having very high similarity to the idealized structure. The analog nature of these scoring functions allows us to compare Q-length dependent trends with observed experimental results. The secondary structure scores for α -helices and β -sheets within polyQ domains of all Q-lengths are shown in Figure 1B. Both α -helix and β -sheet contents increase with Q-length inside the polyQ region. The increase in α -helix content is consistent with the recent time-resolved fluorescence energy transfer (TR-FRET) and electron paramagnetic resonance (EPR) data from Langen and co-workers.¹⁷ Figure 1B shows a monotonic increase α -helix content with Q-length that is similar to experimentally observed trends.^{26,29} The observed increase in β -sheet content with

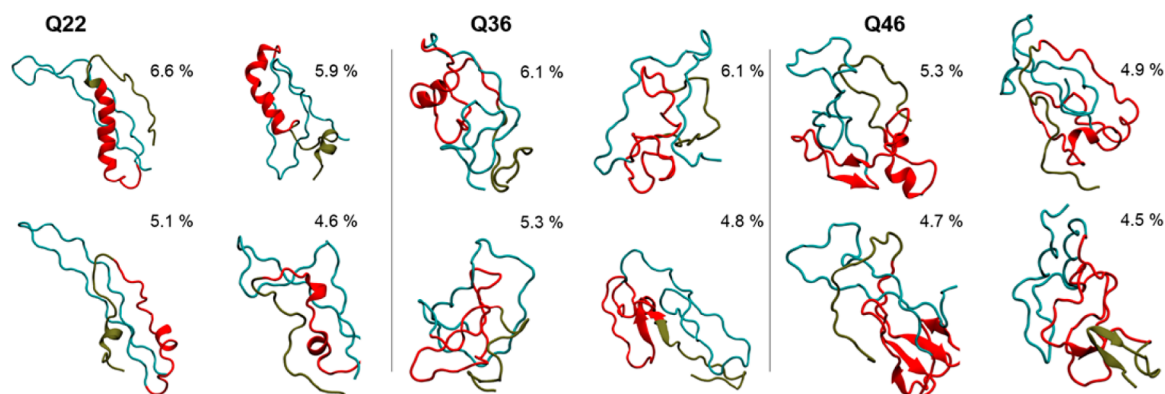


Figure 2. Representative structures determined from clustering analysis of Q22, Q36, and Q46 data sets. The populations of each cluster, as a percentage from sampled conformations, is shown above each structure. In general, as the Q-length increases, the β -sheet content of the polyQ region increases. The N17 domain can also form more β -sheet structures as the Q-length increases. The N17 region is shown in tan, the polyQ in red, and the C38 in cyan. The proteins are rendered using the New Cartoon representation.

increasing Q-length, however, is greater than the increase in α -helix content, especially between Q22 and Q36. This rise in β content has been seen in polyQ studies that found that increasing Q-lengths lead to increases in β -sheet transitions.^{14,26} In particular, the monotonic increase in secondary structure content is manifested only when using a continuous measure of secondary structure (Figure 1B) and not a binary secondary structure measure (Figure 1A). This indicates that the emerging α -helix and β -sheet content in longer Q-lengths is transient rather than static.

By contrast, we find that secondary structural motifs within N17 and C38 do not change significantly with increasing Q-length. Figure 1B illustrates that both α -helix and β -sheet scores for N17 are low in the wild type and remain so in the mtHtt. These results suggest that the N17 domain of the exon-1 monomer is mostly disordered in solution, an observation consistent with previous NMR measurements.¹⁰ Compared to N17, C38 is relatively rigid due to the polyproline chains that dominate its sequence. We assigned the left-handed polyproline II helix (PPII) designation to any two sequential proline residues within C38 that adopt backbone dihedral angles (φ , ψ) inside the range ($-75^\circ \pm 15^\circ$, $150^\circ \pm 15^\circ$). C38's polyproline chains assume PPII secondary structures to a large and effectively constant extent among wt and mtHtt (Figure 1C),³⁰ a result also consistent with previous experiments.³¹

As noted above, β -sheet rich conformations, in particular, become more prevalent as Q-length increases. To further illustrate this trend, Figure 2 shows high-probability structures derived from clustering analyses of three representative polyQ domains: Q22 (wtHtt), Q36 (mtHtt at the pathogenic threshold), and Q46 (mtHtt far above the pathogenic threshold). Clusters were derived from the GROMACS 5.0.5 cluster analysis tool³² based on the algorithm developed by Daura et al.,³³ applying a 3.5 Å RMSD cutoff. The images highlighted in Figure 2 represent the four most populated clusters within each Q-length's data set. The wild type Q22 exhibits more propensity to form helical structures, although these structures are transient, with the majority of sampled structures being disordered. A recent solution NMR study found a propensity to form helical structures in an N17Q17P6 model of Htt exon-1 at low pH and disorder protein behavior at neutral pH.³⁴ Additionally, we found that the Q22 variant features more elongated C38 configurations than seen in Q36 and Q46. The structures of Q36 are the most disordered

among the three Q-lengths, though some β -sheet motifs do start to appear within its polyQ domain. Q46 configurations, however, feature pronounced β -sheet content, particularly within the third cluster wherein four β -strands coalesce inside the polyQ tract. These types of β -sheet conformations may represent structural motifs capable of seeding the oligomerization processes seen in some polyQ aggregation studies.^{14,26,27} Although interactions between helical N17 have been shown to be an important nucleation event in Htt exon-1 oligomerization and may be the most kinetically favorable process,^{29,35} competing pathways have been suggested where the Htt exon-1 monomer misfolds into a β -sheet conformation that then aggregates via β -sheet interactions.^{35–37} Although these β -sheet dependent oligomerization pathways may be much less kinetically favorable, they may contribute to the polymorphic nature of the Htt exon-1 aggregates and may play an important role in its toxicity.

Intra- and Interdomain Tertiary Contacts. In order to characterize the tertiary contacts between the different domains of the exon-1 monomer, we calculated the average number of hydrogen bonds occurring among all residue pairs (Figure 3A) and in summation within and among various domains (Figure 3B). In general, we did not see any persistent contacts among individual residues (Figure 3A). When examining the average total numbers of hydrogen bonds, however, the polyglutamine region displays an unusually high hydrogen bonding propensity in relation to typical proteins (Figure 3B, left panel). To conduct this comparison, we calculated the total number of hydrogen bonds within all standard globular α/β proteins in the Protein Data Bank (PDB), limiting our computation to structures that have at least 20% α -helical and 20% β -sheet content. Strikingly, the number of hydrogen bonds within the polyQ region of exon-1 increases much more rapidly than it does in globular proteins, as a function of chain length (Figure 3B, left panel). These greatly enhanced hydrogen bonding characteristics likely result from a higher propensity for glutamine residues to form hydrogen bonds among themselves rather than with water. The mid panel of Figure 3B shows that side chains of the polyQ form more side chain-side chain and side chain-backbone hydrogen bonds than general structured proteins with comparable sizes from PDB. Interestingly, the number of side chain-backbone interactions within the polyQ follows a trend similar to the number of backbone hydrogen bonds found in structured proteins. This observation

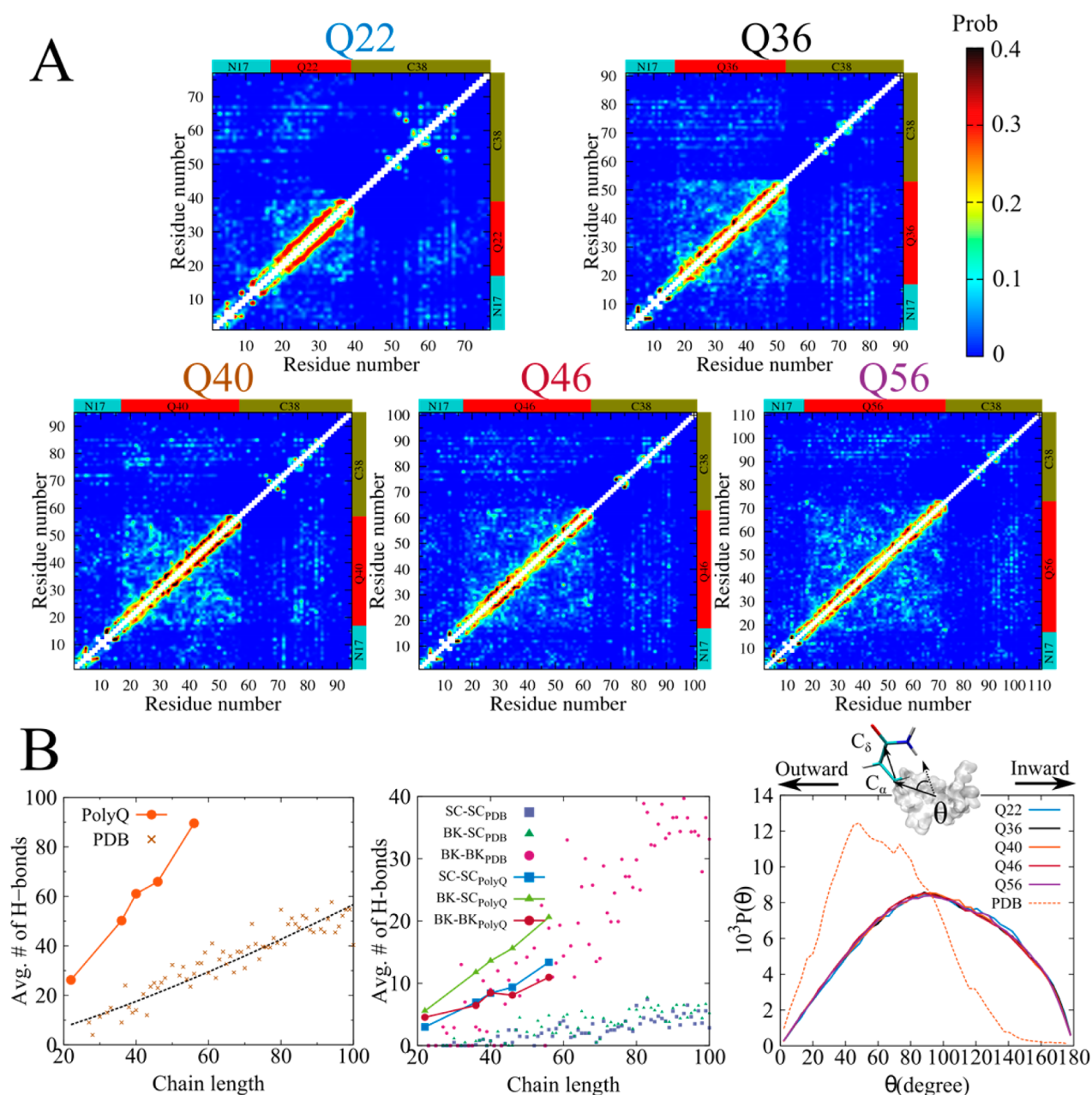


Figure 3. Hydrogen bond analysis shows that the contacts between N17 and C38 are reduced as the Q-length is increased. Panel (A) Pairwise residue hydrogen bonding probability map for all Q-lengths. Panel (B) (Left) Average number of hydrogen bonds in the polyQ region for all Q-lengths (orange circles). For comparison, the average numbers of hydrogen bonds in globular proteins selected from PDB structures are also plotted (brown crosses). The dashed line is a power-law fit with the scaling exponent of 1.28. As seen in the figure, the number of hydrogen bonds within the polyQ region increases at a faster rate than in other globular proteins. (Middle) Average number of side chain-side chain (blue square), backbone-side chain (green triangle) and backbone–backbone (red circle) hydrogen bonds for polyQ and PDB proteins. The side chains of polyQ show more hydrogen-bonding propensity than the PDB proteins with comparable sizes. (Right) Side chain orientation with respect to protein surface. θ is an angle between a vector from center of mass of a protein to C_α atom and another vector from C_α to C_δ . Side chain of GLN in polyQ tends to point inward as compared to general PDB proteins, suggesting that the GLN residues will not be available to form favorable interactions with the solvent.

demonstrates glutamine side chain's “glue”-like property (glutamine acting as “glue”), caused by strong hydrogen-bonding tendency with backbones as well as other GLN side chains. Furthermore, when only buried glutamines are considered, defined as amino acids having less than 4 water molecules within 3 Å, the increase in side chain-side chain hydrogen bonds becomes even more prominent (Figure S1), which implies strong intrahydrogen bonding propensity of polyQ. Because of this strong intrahydrogen bonding, the side chains of glutamine residues are more likely to point inward to

the center of polyQ. In contrast, in general structured proteins, the side chains of glutamine residues mostly point outward (right panel of Figure 3B). To further illustrate this point, we analyzed the average number of side chain-water and side chain-protein hydrogen bonds for surface GLN residues in polyQ and in globular proteins (Table S1). Surface GLN residues were defined as having at least 6 water molecules within 3 Å of the residue. As expected, we did find that polyQ surface GLN residues have more side chain-protein hydrogen bonds than the globular proteins; while the side chain-water

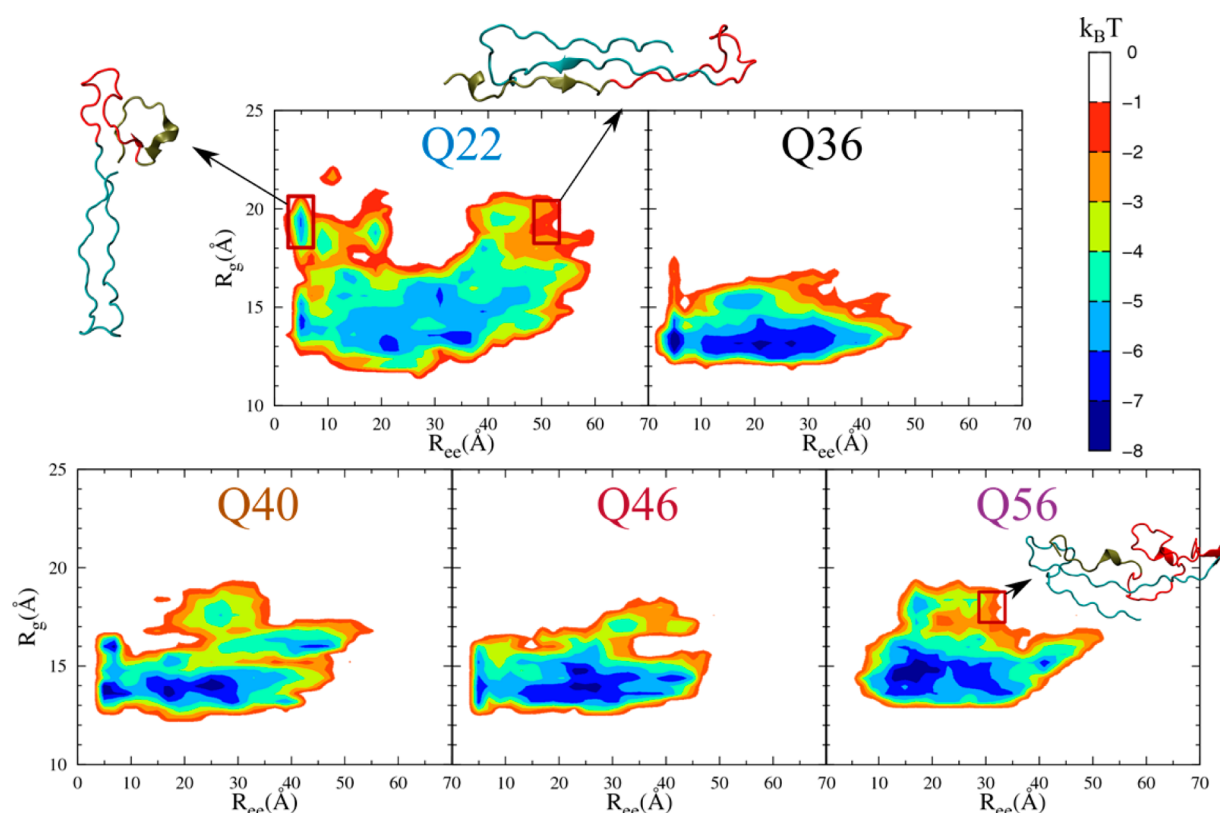


Figure 4. Two-dimensional PMFs projected onto R_{ee} and R_g for the full exon-1 across all Q-lengths. The PMFs are plotted as differences from a global PMF maximum. Q22 has much wider distributions of R_g and R_{ee} values than the other Q-lengths, indicating a more flexible structure. The reduction in the sampled R_g and R_{ee} regions indicates that increases in Q-length lead to more compact structures. For Q22, the overlaid representative structures from the highlighted regions show that the maximum R_g values can be sampled from both the minimum and maximum R_{ee} values, indicating that the extended C38 region is very flexible. On the other hand, the representative structure highlighted in Q56 shows that the protein is relatively compact with the end-end distance less variant even for maximum R_g structures.

hydrogen bonds are significantly less, approximately 1.3 less on average, as compared to globular proteins (Table S1). This suggests that surface GLN residues in polyQ have a lower propensity to interact with water and a higher propensity to face inward than GLN residues in globular proteins. This helps to explain the insolubility of polyQ chains without flanking domains. The less “intrusive” surface GLN side chains to the first solvation shell are also beneficial to the overall folding free energy due to the less disruption of water hydrogen bonding network by the GLN planar side chain amide groups.^{38,39}

Figure S2A shows the average total numbers of hydrogen bonds between N17 and polyQ and C38 and polyQ. Both sets of interdomain hydrogen bonds also increase as Q-length increases, indicating augmented interactions between N17 and C38 with the central polyQ domain. Interestingly, hydrogen bonding between the N17 and C38 regions actually decreases to a small extent as Q-length increases (Figure S2B).

Recent FRET experiments by Caron et al. indicated that the N17 and C38 domains of exon-1 are spatially proximate in Htt of $Q < 36$ but distant in mutants at $Q \geq 36$.²⁵ As a result, Caron et al. surmised that the polyQ region in wtHtt favors disordered states whereas the polyQ in mtHtt tends to form β -sheets. Such a secondary structural shift in polyQ should result in a decrease in contacts between N17 and C38, as we indeed observed in our MD data. To further confirm this correlation, we plotted 2D histograms of N17–C38 hydrogen bond counts against α -helix or β -sheet scores for the polyQ region (Figure S3), and we estimated correlation coefficients using simple

linear regression. These calculations clearly demonstrate that the number of hydrogen bonds between N17 and C38 is inversely correlated with β -sheet content in the polyQ region (Figure S3A). The negative correlation coefficients between β -sheet content and N17–C38 hydrogen bonding, though moderate, seem to generally increase with Q-length (Q56 being a notable exception). By contrast, α -helix scores show very little correlation with hydrogen bonding between N17 and C38 (Figure S3B).

Wild-Type Htt Exhibits More Diverse Conformations than Mutant Htt. We also computed potentials of mean force (PMFs) for the full exon-1 monomer at all Q-lengths, projected onto radius of gyration (R_g) and end-to-end distance (R_{ee}) order parameters (Figure 4). Specifically, the PMFs were calculated as $F(R_g, R_{ee}) = -k_B T [\ln P(R_g, R_{ee}) - \ln P_{\min}(R_g, R_{ee})]$, where k_B is the Boltzmann constant, T is room temperature, $P(R_g, R_{ee})$ is the probability of a given (R_g, R_{ee}) bin, and $P_{\min}$ is the nonzero minimum value in the 2D probability distribution. Put in simple terms, the PMF for Q22 covers a much wider area than any of the other PMFs corresponding to mtHtt. This wide range is partially due to Q22 being able to sample structures that include an extended C38 domain (Figure 4, Q22), which is not seen in longer Q-lengths (Figure 4, Q56). This observation also suggests that wtHtt is more flexible and adopts a more diverse set of conformations than the longer polyQ mutants, which favor more compact structures.

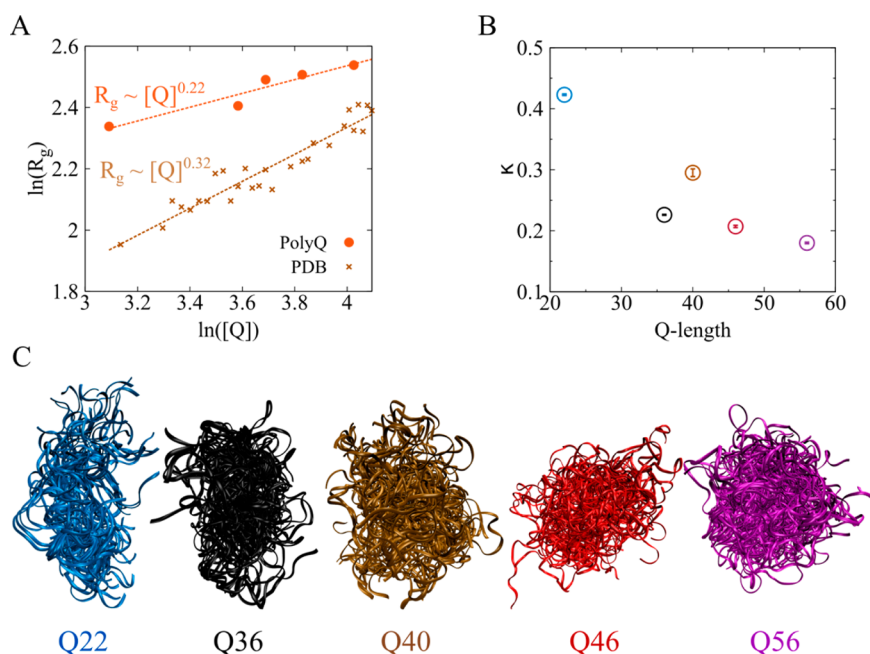


Figure 5. Increasing Q-length leads to supercompact conformations in the polyQ region. (A) Radii of gyration for the polyQ region (orange $\odot$) and structured proteins from the PDB (brown $\times$), as a function of chain length. The dashed lines indicate power-law fitting. The scaling exponents are displayed in the figure, showing that the polyQ will adopt supercompact structures at longer Q-lengths. (B) Relative anisotropy, κ , for all Q-lengths. κ roughly decreases as Q-length increases except for Q40, which indicates that the chain becomes more spherical. (C) Ensemble of polyQ structures. 40 randomly selected polyQ structures are rendered together to illustrate the morphological change in polyQ with increasing Q-length. Q22 has a more aspherical shape than the other mtHtts, and the shape of the protein becomes more spherical as Q-length increases.

Scaling Behavior of Polyglutamines. Intrigued by this striking compactness in mtHtts, we next investigated the scaling behavior of polyQ packing density as a function of Q-length. For purposes of comparison, we calculated the average R_g as a function of chain length, for both polyQ domains studied here and globular protein structures taken from the PDB. Interestingly, we find that as Q-length increases, the compactness of each polyQ domain (as evaluated by R_g) increases at a much faster rate than expected for standard proteins or polymers (Figure 5A). The power-law scaling exponent for the R_g of a structured globular protein is expected to be 0.33,⁴⁰ while the scaling exponent for a relatively unfolded (well-solvated) protein is expected to be ~ 0.5 (0.6 for an ideal polymer in good solvent).^{41,42} Our calculation shows that the proteins from PDB meet these theoretical expectations (Figure 5A, dashed brown line); however, the scaling exponent for polyQ is estimated at a surprisingly low 0.22, despite that glutamine itself is hydrophilic. These scaling data indicate a Q-length dependent supercompactness in polyQ, a phenomenon that most likely results from the strong internal hydrogen bonding between glutamine residues (Figure 3B).

Along with this supercompactness, we also find that the shape of the polyQ tract becomes more spherical as the Q-length increases. In order to assess such a change, we estimated the relative anisotropy parameter, κ , which measures the shape anisotropy of a polymer. For perfect spheres, $\kappa = 0$; for a perfectly straight polymer, $\kappa = 1$. Figure 5B shows that κ generally decreases with increasing Q-length, indicating that the polyQ changes from a rod-like morphology to a spherical shape at longer Q-lengths. The polyQ structural ensembles shown in Figure 5C—each generated from 40 random snapshots—illustrate this trend.

DISCUSSION

In this work, we performed the first systematic, all-atom MD study of the full Huntingtin exon-1 protein across five different polyglutamine segment lengths. These simulations have provided new insights into the effects of elongated polyQ tracts on Htt's structure, hinting at the molecular basis for the relationship between CAG repeats and Huntington's disease pathology. As Q-length is increased, the Htt exon-1 proteins exhibit more β -sheet content in general, which acts to reduce contact between the flanking N17 and C38 domains and results in enhanced compactness in the polyQ region. As such, the toxicity of the elongated polyQ tract can perhaps be partially attributed to two sources: aggregation that is aided by an increase β -sheet propensity and an increase in compactness that likely changes the ways in which Htt interacts with itself, other Htts, and distinct protein binding partners.

Our clustering analysis highlights the emergence of β -sheet rich conformations with increasing Q-length. We find that the Q22 polyQ tract does not contain any β -sheet content in its most populous clusters, instead favoring very helical structures. Contrastingly, the elongated polyQ tracts (especially Q46) feature structures that are rich in β -sheet content. Such β -sheet rich conformations are typical of the motifs sometimes associated with oligomer and fibril formation, suggesting that these configurations might contribute to the increased aggregation of the mtHtt.

Our findings also demonstrate how the elongation of the polyQ segment alters interactions between the N17 and C38 domains. While the number of hydrogen bonds within the polyQ region itself increases with Q-length (in fact, at a much faster rate than expected for globular proteins), the number of hydrogen bonds between N17 and C38 decreases with increasing Q-length. The reduction in N17–C38 hydrogen

bonds is correlated with an increase of β -sheet structures in the polyQ region. This observation confirms previously reported experimental FRET results and a proposed model by Truant and co-workers.²⁵

As the polyQ-length increases, the overall conformational flexibility of the protein is diminished, and the exon-1 mutants adopt increasingly compact structures. Along with the appearance of β -sheet rich conformations, increased compactness and reduced flexibility appear to represent key differences between the wild type and pathogenic Htt variants. It is important to emphasize, however, that even though extending the Q-length leads to reduced conformational flexibility, the exon-1 protein remains disordered and does not adopt a native folded state.

Ultimately, this loss of flexibility stimulates a surprising finding: the radial scaling behavior of polyQ domains is very different from that of standard globular proteins, such that the polyQ region adopts supercompact structures at longer Q-lengths. The R_g scaling exponent for polyQ was estimated to be 0.22, a considerably smaller value than the exponent typical of globular proteins (0.33). Supercompactness is an astonishing characteristic for an IDP to possess: based on conventional wisdom, one expects IDPs to abide by a scaling exponent of approximately ~ 0.5 . This unanticipated scaling behavior likely results from the explosion in internal hydrogen bonding seen with increasing Q-length. The resultant reduction in flexibility and supercompactness should impact the way individual mtHttS interact with other proteins, possibly inducing a loss of function and thereby accessing an orthogonal mechanism of toxicity. One possibility is that the enhanced compactness and resulting mechanical stability may alter the proteolysis mechanism of mtHttS (e.g., exceeding the capability of proteasome to digest them), thus suppressing the degradation of these mutant proteins and leading to their accumulation in vivo.⁴³

Although our findings suggest these possible origins for the toxicity of mtHttS, further work is needed to discover just how these changes in structure affect protein–protein interactions and what roles the N17 and C38 regions play in mediating those downstream effects. Additionally, comparison of Htt with other disease-causing polyglutamine proteins may provide useful insights into the role of the flanking domains play in modulating the pathogenic Q-length. While a long road lies ahead with respect to comprehending the molecular underpinnings of Huntington's disease, the Q-length-dependent perturbations to Huntingtin exon-1 monomer structure identified here represent encouraging early steps toward that direction.

MATERIALS AND METHODS

Molecular dynamics simulations were conducted on Htt exon-1 proteins (MATLEKLMKAFESLKFSQ_nP₁₁QLPQP₃QAQPLL-PQPQP₁₀) featuring five different Q-lengths: Q22, Q36, Q40, Q46, and Q56. All simulated exon-1 proteins included both the N-terminal (N17) and C-terminal (C38) domains flanking the central polyQ tract. Due to the intrinsic disorder of the Htt exon-1, we used temperature replica exchange molecular dynamics (T-REMD)⁴⁴ to enhance the sampling of the conformational free energy surface. Furthermore, to remove biases from any known structural motif, we prepared initial structures for T-REMD using the following procedure:^{45–47} polypeptide chains with fully extended conformations were constructed, collapsed in vacuo through minimization and equilibration at 700 K, solvated with a 2.0 nm buffer of TIP3P water, and neutralized with NaCl to a concentration of 100 mM. To obtain fully randomized

conformations, the system was heated from 298 to 600 K and maintained at that high temperature for 400 ns. Over the final 128 ns of this trajectory, 128 initial conformations for T-REMD seeding were selected at a rate of one per 1 ns.

After a 1 ns equilibration, we carried out standard T-REMD simulations with 128 replicas at constant volume for ca. 500 ns per replica, using temperature ranges from 298 to 600 K. The replicas were allowed to exchange temperatures every 1000 time steps and the resulting acceptance ratio was in the range of 0.25–0.3. A time step of 4 fs with virtual sites was used. All simulations were performed with GROMACS 5.0.5³² using the OPLS-AA/L force field.⁴⁸

The total aggregate simulation time for our data set reached 320 μ s. We recorded atomic coordinates at every 0.1 ns and analyzed the trajectories at temperatures ranging from 298 to 315 K, after discarding the first 100 ns of each run. Secondary structure analyses were carried out using STRIDE⁴⁹ and the PLUMED software package.³⁰ Convergence checks (Figure S4–S5) and further analysis can be found in the Supporting Information. Molecular structures were visualized and rendered using VMD⁵¹ (<http://www.ks.uiuc.edu/Research/vmd/>)

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.7b00838.

Figure S1–S5; Table S1 (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*ruhongz@us.ibm.com

ORCID

Payel Das: 0000-0002-7288-0516

Ruhong Zhou: 0000-0001-8624-5591

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We would like to thank Jeffrey K. Weber, Tien Huynh, Thomas F. Vogt, Amrita Mohan, Seung-gu Kang, Joseph Morrone and Alicia Preiss for their various help with this work. This work was partially funded by CHDI Foundation Inc., a charitable foundation that funds research into Huntington's disease. ML acknowledges the support of NIH (GM115749). RZ acknowledges the support of the IBM Blue Gene Science Program (W1258591, W1464125, W1464164).

■ REFERENCES

- (1) MacDonald, M. E.; Ambrose, C. M.; Duyao, M. P.; Myers, R. H.; Lin, C.; Srinidhi, L.; Barnes, G.; Taylor, S. A.; James, M.; Groot, N.; MacFarlane, H.; Jenkins, B.; Anderson, M. A.; Wexler, N. S.; Gusella, J. F.; Bates, G. P.; Baxendale, S.; Hummerich, H.; Kirby, S.; North, M.; Youngman, S.; Mott, R.; Zehetner, G.; Sedlacek, Z.; Poustka, A.; Frischauf, A.-M.; Lehrach, H.; Buckler, A. J.; Church, D.; Doucette-Stamm, L.; O'Donovan, M. C.; Riba-Ramirez, L.; Shah, M.; Stanton, V. P.; Strobel, S. A.; Draths, K. M.; Wales, J. L.; Dervan, P.; Housman, D. E.; Altherr, M.; Shiang, R.; Thompson, L.; Fielder, T.; Wasmuth, J. J.; Tagle, D.; Valdes, J.; Elmer, L.; Allard, M.; Castilla, L.; Swaroop, M.; Blanchard, K.; Collins, F. S.; Snell, R.; Holloway, T.; Gillespie, K.; Datson, N.; Shaw, D.; Harper, P. S. *Cell* **1993**, 72 (6), 971–983.
- (2) Wexler, N. S. *Proc. Natl. Acad. Sci. U. S. A.* **2004**, 101, 3498–3503.
- (3) Gray, M.; Shirasaki, D. I.; Cepeda, C.; Andre, V. M.; Wilburn, B.; Lu, X.-H.; Tao, J.; Yamazaki, I.; Li, S.-H.; Sun, Y. E.; Li, X.-J.; Levine, M. S.; Yang, X. W. *J. Neurosci.* **2008**, 28 (24), 6182–6195.
- (4) Saudou, F.; Finkbeiner, S.; Devys, D.; Greenberg, M. E. *Cell* **1998**, 95 (1), 55–66.

- (5) Perutz, M. F.; Johnson, T.; Suzuki, M.; Finch, J. T. *Proc. Natl. Acad. Sci. U. S. A.* **1994**, *91*, 5355–5358.
- (6) Miller, J.; Arrasate, M.; Brooks, E.; Libeu, C. P.; Legleiter, J.; Hatters, D.; Curtis, J.; Cheung, K.; Krishnan, P.; Mitra, S.; Widjaja, K.; Shaby, B. A.; Lotz, G. P.; Newhouse, Y.; Mitchell, E. J.; Osmand, A.; Gray, M.; Thulasiramin, V.; Frederic, Segal, M.; Yang, X. W.; Masliah, E.; Thompson, L. M.; Muchowski, P. J.; Weisgraber, K. H.; Finkbeiner, S. *Nat. Chem. Biol.* **2011**, *7*, 925–934.
- (7) Kim, M. W.; Chelliah, Y.; Kim, S. W.; Otwinowski, Z.; Bezprozvanny, I. *Structure* **2009**, *17* (9), 1205–1212.
- (8) Kim, M. *Prion* **2013**, *7* (3), 221–228.
- (9) Michalek, M.; Salnikov, E. S.; Bechinger, B. *Biophys. J.* **2013**, *105* (3), 699–710.
- (10) Thakur, A. K.; Jayaraman, M.; Mishra, R.; Thakur, M.; Chellgren, V. M.; Byeon, I.-J. L.; Anjum, D. H.; Kodali, R.; Creamer, T. P.; Conway, J. F.; Gronenborn, A. M.; Wetzel, R. *Nat. Struct. Mol. Biol.* **2009**, *16* (3), 380–389.
- (11) Sahoo, B.; Singer, D.; Kodali, R.; Zuchner, T.; Wetzel, R. *Biochemistry* **2014**, *53* (24), 3897–3907.
- (12) Crick, S. L.; Jayaraman, M.; Frieden, C.; Wetzel, R.; Pappu, R. V. *Proc. Natl. Acad. Sci. U. S. A.* **2006**, *103* (45), 16764–16769.
- (13) Miettinen, M. S.; Knecht, V.; Monticelli, L.; Ignatova, Z. *J. Phys. Chem. B* **2012**, *116* (34), 10259–10265.
- (14) Heck, B. S.; Doll, F.; Hauser, K. *Biophys. Chem.* **2014**, *185*, 47–57.
- (15) Hoop, C. L.; Lin, H.-K.; Kar, K.; Magyarfalvi, G.; Lamley, J. M.; Boatz, J. C.; Mandal, A.; Lewandowski, J. R.; Wetzel, R.; van der Wel, P. C. A. *Proc. Natl. Acad. Sci. U. S. A.* **2016**, *113* (6), 1546–1551.
- (16) Chen, S.; Ferrone, F. A.; Wetzel, R. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99* (18), 11884–11889.
- (17) Fodale, V.; Kegulian, N. C.; Verani, M.; Cariulo, C.; Azzollini, L.; Petricca, L.; Daldin, M.; Boggio, R.; Padova, A.; Kuhn, R.; Pacifici, R.; Macdonald, D.; Schoenfeld, R. C.; Park, H.; Isas, J. M.; Langen, R.; Weiss, A.; Caricasole, A. *PLoS One* **2014**, *9* (12), e112262.
- (18) Das, P.; King, J. A.; Zhou, R. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108* (26), 10514–10519.
- (19) Zhou, R. *Proc. Natl. Acad. Sci. U. S. A.* **2003**, *100* (23), 13280–13285.
- (20) Xia, Z.; Yang, Z.; Huynh, T.; King, J. A.; Zhou, R. *Sci. Rep.* **2013**, *3*, 1560.
- (21) Kelley, N. W.; Huang, X.; Tam, S.; Spiess, C.; Frydman, J.; Pande, V. S. *J. Mol. Biol.* **2009**, *388*, 919–927.
- (22) Wang, Y.; Voth, G. A. J. *Phys. Chem. B* **2010**, *114* (26), 8735–8743.
- (23) Długosz, M.; Trylska, J. *J. Phys. Chem. B* **2011**, *115*, 11597–11608.
- (24) Lakhani, V. V.; Ding, F.; Dokholyan, N. V. *PLoS Comput. Biol.* **2010**, *6* (4), e1000772.
- (25) Caron, N. S.; Desmond, C. R.; Xia, J.; Truant, R. *Proc. Natl. Acad. Sci. U. S. A.* **2013**, *110* (36), 14610–14615.
- (26) Nagai, Y.; Inui, T.; Popiel, H. A.; Fujikake, N.; Hasegawa, K.; Urade, Y.; Goto, Y.; Naiki, H.; Toda, T. *Nat. Struct. Mol. Biol.* **2007**, *14* (4), 332–340.
- (27) Kar, K.; Hoop, C. L.; Drombosky, K. W.; Baker, M. A.; Kodali, R.; Arduini, I.; van der Wel, P. C. A.; Horne, W. S.; Wetzel, R. *J. Mol. Biol.* **2013**, *425* (7), 1183–1197.
- (28) Pietrucci, F.; Laio, A. *J. Chem. Theory Comput.* **2009**, *5* (9), 2197–2201.
- (29) Jayaraman, M.; Kodali, R.; Sahoo, B.; Thakur, A. K.; Mayasundari, A.; Mishra, R.; Peterson, C. B.; Wetzel, R. *J. Mol. Biol.* **2012**, *415* (5), 881–899.
- (30) Stapley, B. J.; Creamer, T. P. *Protein Sci.* **1999**, *8* (3), 587–595.
- (31) Darnell, G. D.; Derryberry, J.; Kurutz, J. W.; Meredith, S. C. *Biophys. J.* **2009**, *97* (8), 2295–2305.
- (32) Berendsen, H. J. C.; van der Spoel, D.; van Drunen, R. *Comput. Phys. Commun.* **1995**, *91* (1–3), 43–56.
- (33) Daura, X.; Gademann, K.; Jaun, B.; Seebach, D.; van Gunsteren, W. F.; Mark, A. E. *Angew. Chem., Int. Ed.* **1999**, *38* (1–2), 236–240.
- (34) Baias, M.; Smith, P. E. S.; Shen, K.; Joachimiak, L. A.; Žerko, S.; Koźmiński, W.; Frydman, J.; Frydman, L. *J. Am. Chem. Soc.* **2017**, *139* (3), 1168–1176.
- (35) Jayaraman, M.; Mishra, R.; Kodali, R.; Thakur, A. K.; Koharudin, L. M. I.; Gronenborn, A. M.; Wetzel, R. *Biochemistry* **2012**, *51* (13), 2706–2716.
- (36) Kokona, B.; Rosenthal, Z. P.; Fairman, R. *Biochemistry* **2014**, *53* (43), 6738–6746.
- (37) Kokona, B.; Johnson, K. A.; Fairman, R. *Biochemistry* **2014**, *53* (43), 6747–6753.
- (38) Liu, P.; Huang, X.; Zhou, R.; Berne, B. J. *Nature* **2005**, *437* (7055), 159–162.
- (39) Zhou, R.; Huang, X.; Margulis, C. J.; Berne, B. J. *Science* **2004**, *305* (5690), 1605–1609.
- (40) Dima, R. I.; Thirumalai, D. *J. Phys. Chem. B* **2004**, *108* (21), 6564–6570.
- (41) Rawat, N.; Biswas, P. *J. Chem. Phys.* **2009**, *131*, 165104.
- (42) Hofmann, H.; Soranno, A.; Borgia, A.; Gast, K.; Nettels, D.; Schuler, B. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109* (40), 16155–16160.
- (43) Dougan, L.; Li, J.; Badilla, C. L.; Berne, B. J.; Fernandez, J. M. *Proc. Natl. Acad. Sci. U. S. A.* **2009**, *106* (31), 12605–12610.
- (44) Sugita, Y.; Okamoto, Y. *Chem. Phys. Lett.* **1999**, *314* (1–2), 141–151.
- (45) An, D.; Su, J.; Weber, J. K.; Gao, X.; Zhou, R.; Li, J. *J. Am. Chem. Soc.* **2015**, *137* (26), 8412–8418.
- (46) Kang, S. G.; Huynh, T.; Xia, Z.; Zhang, Y.; Fang, H.; Wei, G.; Zhou, R. *J. Am. Chem. Soc.* **2013**, *135* (8), 3150–3157.
- (47) Xia, Z.; Das, P.; Shakhnovich, E. I.; Zhou, R. *J. Am. Chem. Soc.* **2012**, *134*, 18266–18274.
- (48) Kaminski, G. A.; Friesner, a. R. A.; Tirado-Rives, J.; Jorgensen, W. L. *J. Phys. Chem. B* **2001**, *105* (28), 6474–6487.
- (49) Frishman, D.; Argos, P. *Proteins: Struct., Funct., Genet.* **1995**, *23* (4), 566–579.
- (50) Bonomi, M.; Branduardi, D.; Bussi, G.; Camilloni, C.; Provasi, D.; Raiteri, P.; Donadio, D.; Marinelli, F.; Pietrucci, F.; Broglia, R. A.; Parrinello, M. *Comput. Phys. Commun.* **2009**, *180* (10), 1961–1972.
- (51) Humphrey, W.; Dalke, A.; Schulten, K. *J. Mol. Graphics* **1996**, *14* (1), 33–38.